Consultancy Report: Central England Temperature Change

Response to Brief 6 — HadCET Seasonal Analysis (1879–2024)

Contents

1 Introduction **2**

2 Part I: Visual Evidence **2**

2.1 Decision 1 – Is There a Sustained Warming Shift, and How Large Is It? 2

2.2 Decision 2 – Which Seasons Are Changing Most, and What Kind of Change Is It? 3

2.3 Decision 3 – What Does the Distribution Evidence Tell Us About Monitoring? 5

3 Part II: Statistical Modelling **6**

3.1 Baseline Shift Results 7

3.2 Long-Run Warming Rate Results 7

4 Decision Summary **8**

1 Introduction

This report uses the Met Office HadCET seasonal series (v2.1.0.0) to address the three decisions in Brief 6. The dataset covers seasonal mean temperature from 1659 onward and seasonal minimum and maximum from 1878 onward, representing the Midlands region.

The baseline strategy compares two 30-year windows: 1961–1990 (historical) against 1991–2020 (recent). All three metrics are analysed over the common period 1879–2024. Part I uses visualisations to establish the pattern; Part II fits a GLS AR(1) model to add formal uncertainty estimates.

2 Part I: Visual Evidence

2.1 Decision 1 – Is There a Sustained Warming Shift, and How Large Is It?

We approach this in two steps: first checking whether any warming is genuinely persistent over time, then measuring how large the recent step-up is.

To check persistence, at each year we compute the 30-year running average and track how far it sits above where it started in the late 19th century. Any line above zero means conditions are warmer than at the start of the record.

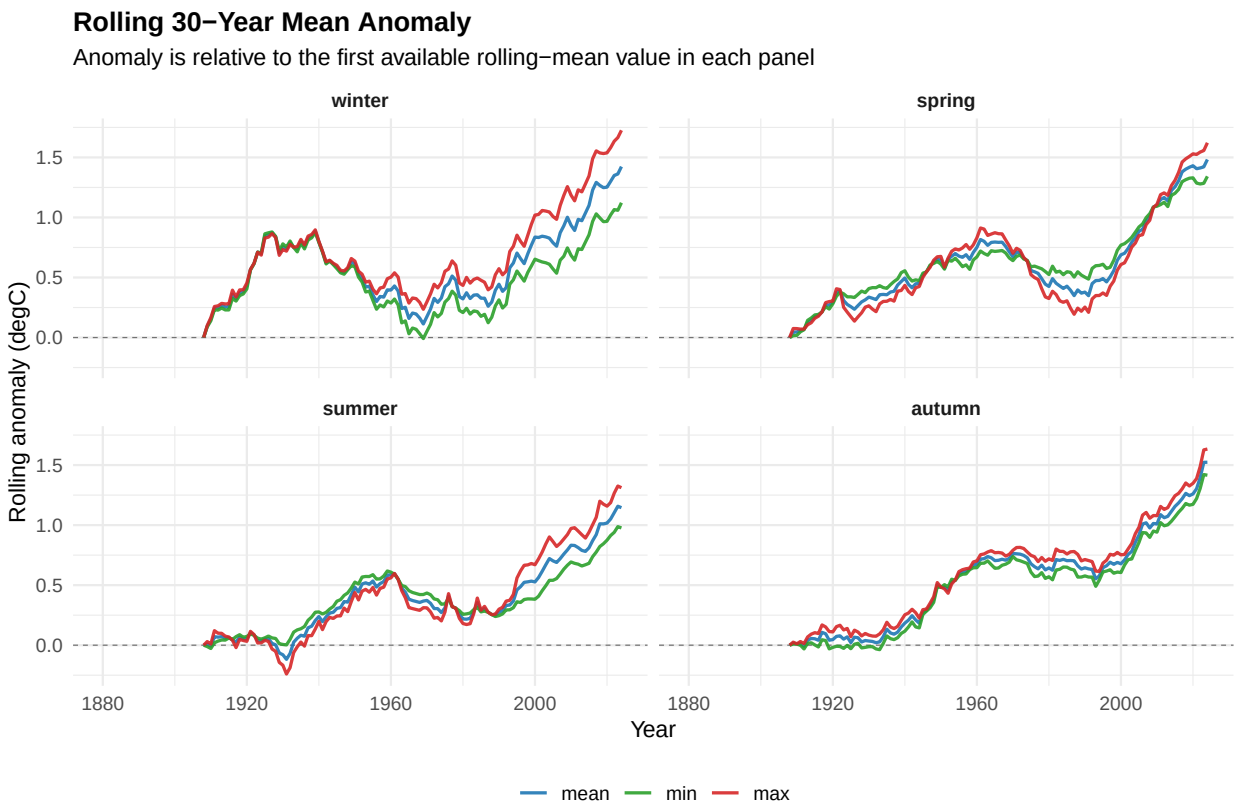

Figure 1: Shift in the running 30-year average relative to the late 19th century baseline, by season and metric. Lines above zero mean the recent average is warmer than at the start of the record.

All four seasons have been above zero since the early 2000s at the latest, and most have been climbing steadily since the 1980s. The consistent upward direction across all 12 panels rules out a single unusual period driving the result.

With persistence confirmed, the next chart measures the step-up directly. Each bar is the 1991–2020 period mean minus the 1961–1990 mean.

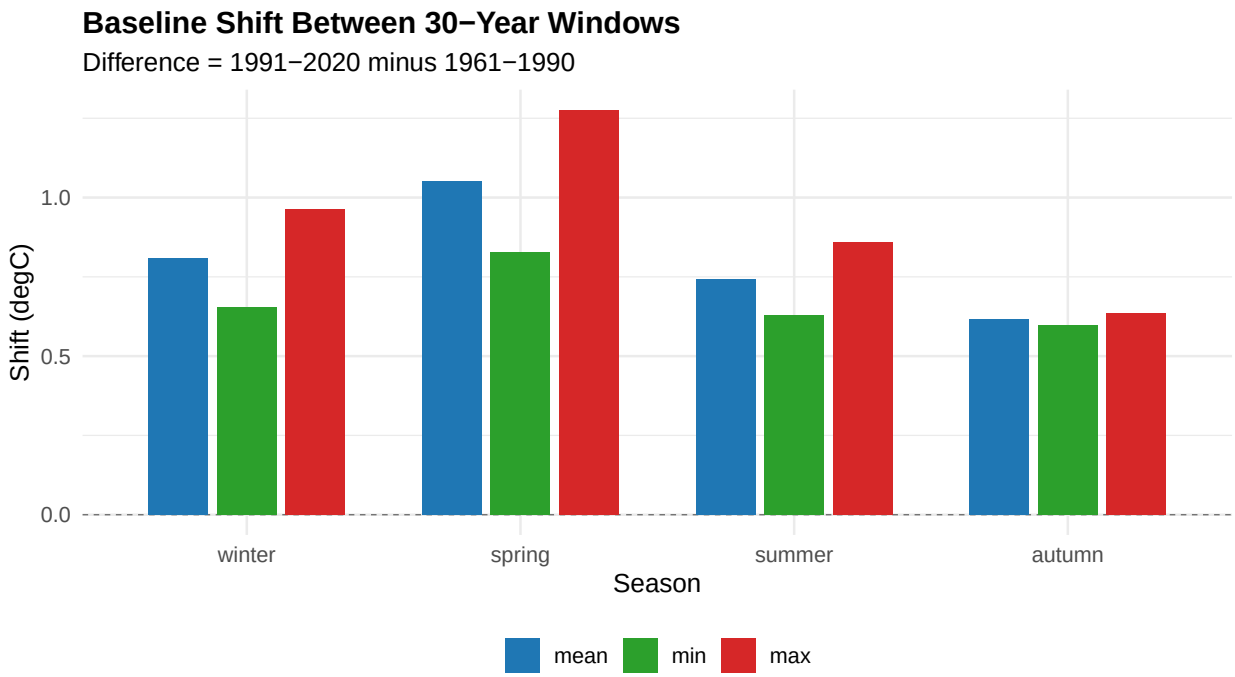

Figure 2: Difference in period mean temperature between 1991–2020 and 1961–1990, by season and metric. Every bar above zero indicates warming.

Every bar is positive. Spring maximum has the largest shift at around +1.28 degC; autumn sits at the lower end at around +0.60 degC. Notably, within every season, maximum temperature shifts more than minimum – meaning warming is more pronounced at the upper end of each season's range than at the lower end. This is worth flagging directly: Brief 6 raised the possibility that minimum temperatures might be rising faster, but the data show the opposite pattern.

2.2 Decision 2 – Which Seasons Are Changing Most, and What Kind of Change Is It?

Decision 2 asks whether the change looks like a gradual trend, a step-change, or shifting variability – three different things for planning purposes.

We start with the full trajectories to see the shape of change before summarising it into rates.

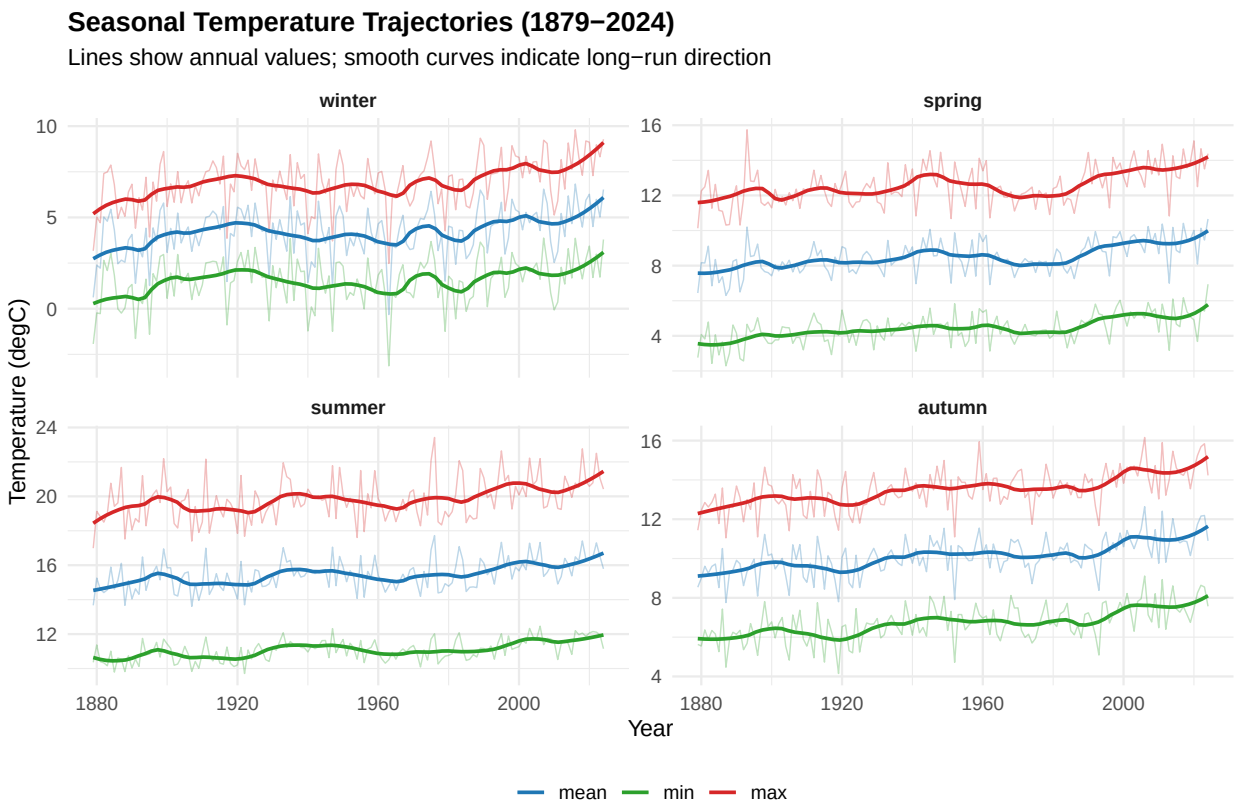

Figure 3: Annual temperature values (faint) and smooth long-run trends (bold) by season and metric, 1879–2024.

All four seasons show upward trends. Winter and summer have a visible mid-century plateau from the 1950s to early 1980s, followed by a sharp upturn; spring and autumn rise more gradually throughout. This isn't a clean linear climb – it's a slow rise that steepened from the late 1980s onward. Formal rate estimates are in Part II.

A warmer average doesn't automatically mean more erratic conditions. The next chart checks whether year-to-year spread has changed as well.

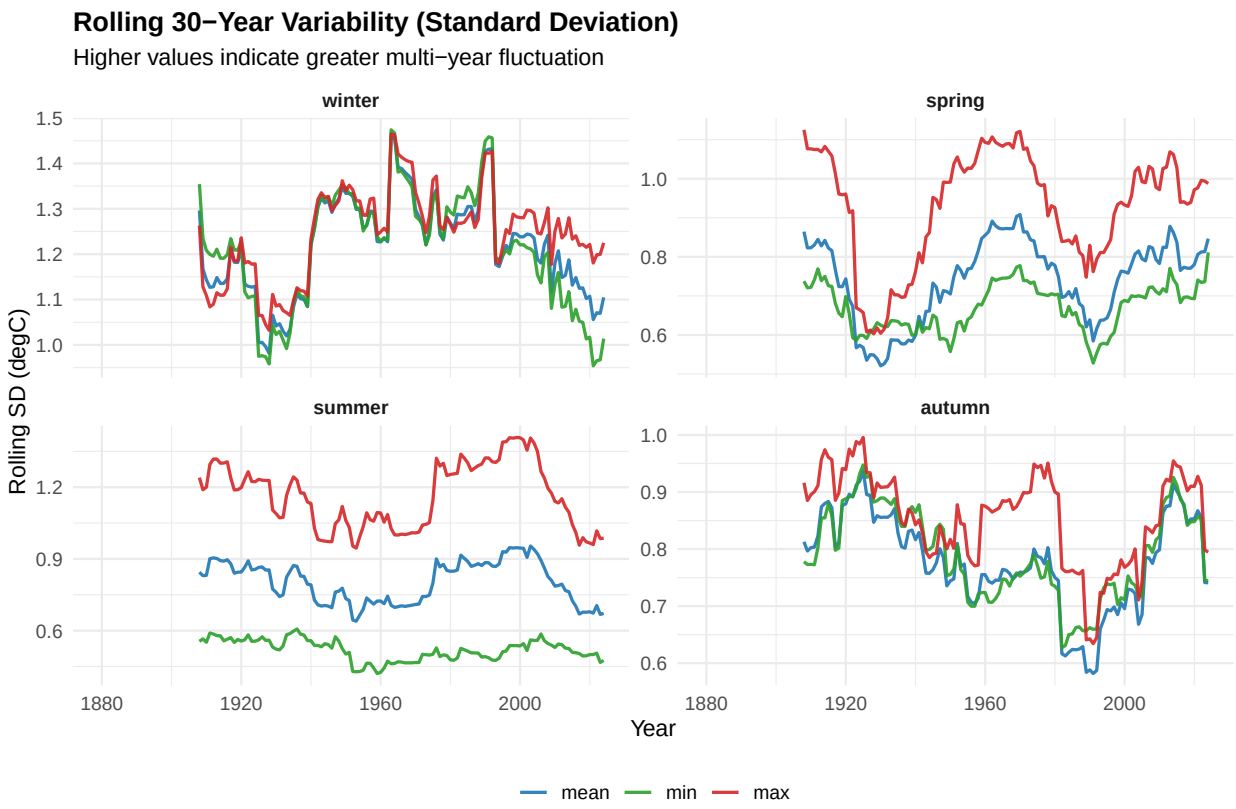

Figure 4: Rolling 30-year standard deviation by season and metric. Rising lines indicate increasing spread; falling lines indicate more consistent conditions.

Winter and summer maximum variability have declined in recent decades – those seasons are warmer on average but also more consistent year to year. Spring and autumn variability is similar to or slightly above the 1961–1990 level. Trend and variability don't move together, so they need to be tracked separately.

2.3 Decision 3 – What Does the Distribution Evidence Tell Us About Monitoring?

For monitoring design, averages and rates alone aren't enough – it helps to see how the full spread of conditions has shifted. The chart below compares the complete distribution in 1961–1990 (blue) and 1991–2020 (orange).

Figure 5: Full temperature distributions for 1961–1990 (blue) and 1991–2020 (orange) by season and metric. Higher position means a warmer distribution; wider shape means greater spread.

Every orange distribution sits higher than the blue. The largest median shifts are in summer maximum and spring maximum, both near or above +1.4 degC. Winter maximum narrows in the recent period while spring and autumn widen slightly, consistent with the variability picture above.

For a practical monitoring framework, three things follow from this. First, mean and maximum should both be reported – minimum alone understates the warming signal. Second, spring and summer maximum are the highest-priority series to watch, given they show the largest distributional shifts and are most relevant to heat-related planning. Third, for defining "material change", a reasonable starting point is to use the 1991–2020 distribution as the current reference baseline and treat a season falling above the 90th percentile of that distribution as a trigger for review. Spring and autumn will need a wider tolerance band than winter and summer, given their higher year-to-year spread in the recent period.

3 Part II: Statistical Modelling

Part I shows clear patterns visually, but can't quantify uncertainty or correct for a structural feature of climate data: nearby years aren't independent. A warm year tends to be followed by another warm year, which makes standard OLS intervals too narrow. We use GLS with AR(1) errors to correct for this, fitting two models per series: a shift model comparing the two 30-year periods, and a trend model estimating the long-run rate over 1879–2024.

3.1 Baseline Shift Results

The table and figure below show the formal AR(1)-corrected shift estimates. Intervals that don't include zero indicate a clear positive shift.

Table 1: GLS AR(1) baseline shift estimates (1991–2020 minus 1961–1990). Cell format is estimate [95% interval] in degC. Asterisk marks intervals that don't include zero.

Season	Mean	Min	Max
Winter	+0.692 [-0.149, +1.534]	+0.540 [-0.283, +1.363]	+0.853 [-0.015, +1.721]
Spring	+1.063 [+0.729, +1.396]*	+0.833 [+0.535, +1.130]*	+1.292 [+0.871, +1.712]*
Summer	+0.716 [+0.221, +1.211]*	+0.620 [+0.316, +0.924]*	+0.808 [+0.069, +1.548]*
Autumn	+0.606 [+0.188, +1.024]*	+0.582 [+0.141, +1.023]*	+0.630 [+0.197, +1.063]*

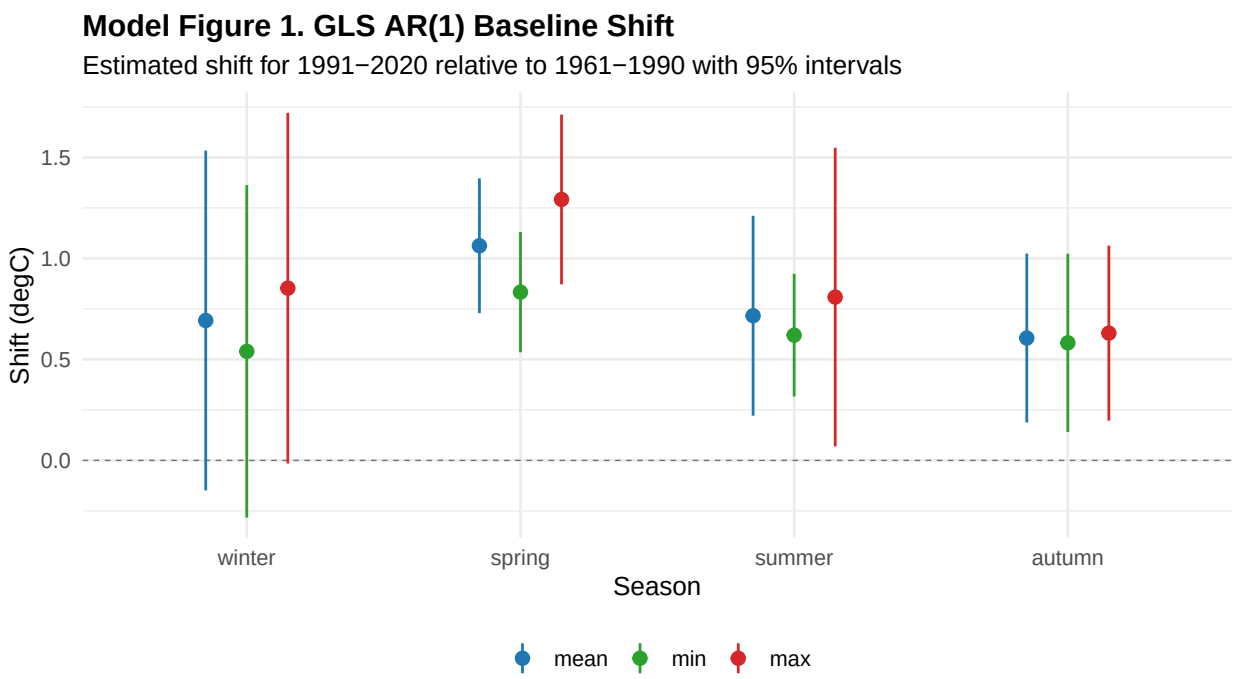

Figure 6: GLS AR(1) baseline shift estimates with 95% intervals. Intervals entirely above zero indicate a clear positive shift.

Spring is the clearest case: all three metrics show intervals entirely above zero, with spring maximum at +1.29 [0.87, 1.71] degC. Summer and autumn are also clear. Winter is positive throughout but intervals include zero for most metrics, reflecting higher year-to-year noise rather than genuine ambiguity about direction.

3.2 Long-Run Warming Rate Results

The trend model uses the full 1879–2024 record rather than comparing two specific windows, making it less sensitive to the choice of baseline years.

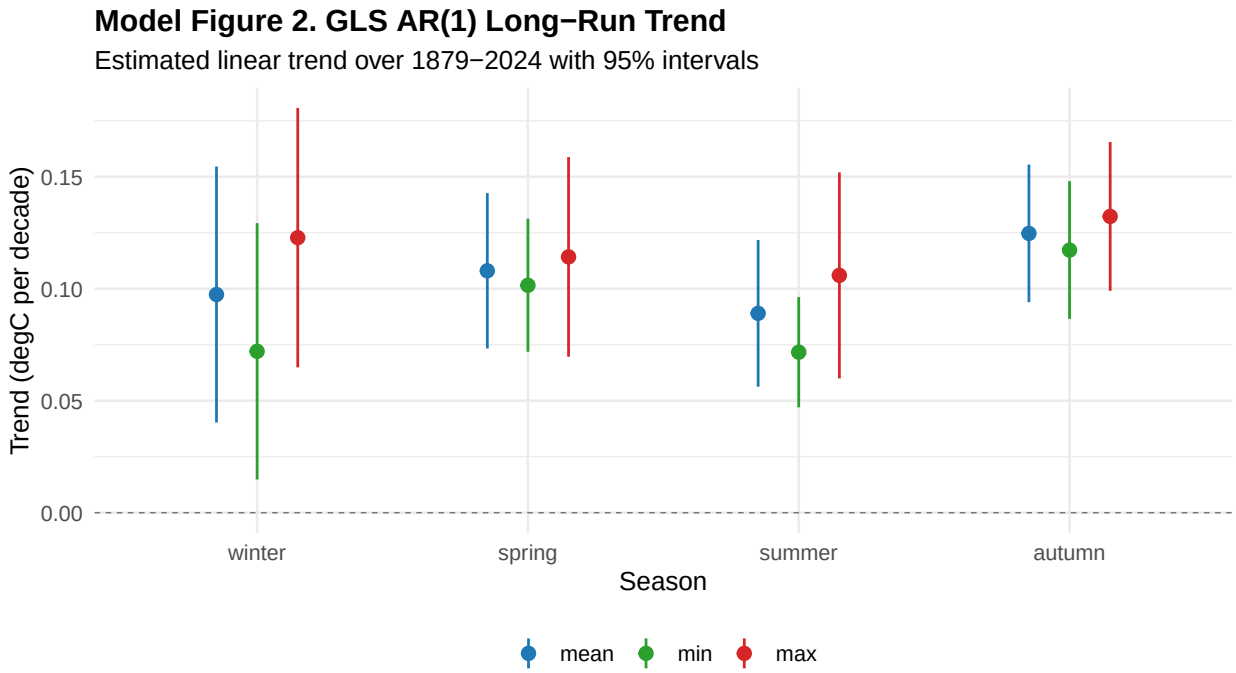

Figure 7: GLS AR(1) long-run warming rate estimates (degC per decade, 1879–2024) with 95% intervals.

Table 2: GLS AR(1) long-run warming rate estimates by season and metric, 1879–2024, in degC per decade. Asterisk marks intervals that don't include zero.

Season	Mean	Min	Max
Winter	+0.097 [+0.040, +0.155]*	+0.072 [+0.015, +0.129]*	+0.123 [+0.065, +0.181]*
Spring	+0.108 [+0.073, +0.143]*	+0.102 [+0.072, +0.131]*	+0.114 [+0.070, +0.159]*
Summer	+0.089 [+0.056, +0.122]*	+0.072 [+0.047, +0.096]*	+0.106 [+0.060, +0.152]*
Autumn	+0.125 [+0.094, +0.155]*	+0.117 [+0.086, +0.148]*	+0.132 [+0.099, +0.166]*

Every one of the 12 series has an interval entirely above zero – including winter, which was less certain in the shift model. Autumn is fastest at +0.12 to +0.13 degC per decade; winter and summer minimum are slowest at around +0.07 to +0.09. The AR(1) correction makes the most difference for winter and summer, where year-to-year correlation is strongest (rho around 0.21–0.22 for winter), widening their intervals relative to a plain OLS approach.

4 Decision Summary

Decision 1. Yes, there is a sustained and material warming shift. All seasons have been above their late 19th century levels since at least the early 2000s, with acceleration from the late 1980s. The step-up is +0.60 to +1.28 degC depending on season and metric. GLS modelling confirms this is statistically clear for spring, summer, and autumn; the long-run trend model confirms it for winter as well.

Decision 2. Spring has the largest recent baseline shift and autumn has the fastest long-run warming rate. The change is a slow rise that steepened from the late 1980s – neither a clean linear trend nor a single step-change. Contrary to the expectation that minimum temperatures might be rising faster, maximum temperature consistently shifts more than minimum across all seasons, so maximum is the more sensitive

indicator. Variability has declined in winter and summer but stayed similar or slightly higher in spring and autumn, so level and variability need to be tracked independently.

Decision 3. We recommend reporting both mean and maximum temperature by season, with minimum as a secondary indicator. Spring and summer maximum are the highest-priority series given their largest distributional shifts. A working definition of material change is a season falling above the 90th percentile of its 1991–2020 distribution; spring and autumn should carry a slightly wider tolerance than winter and summer given their higher recent spread. The 1991–2020 period provides the appropriate reference baseline for all thresholds.